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ORIGINAL RESEARCH

Optimal design and sensitivity analysis of distributed biomass-based hybrid renewable energy systems for rural electrification: Case study of different photovoltaic/wind/battery-integrated options in Babadam, northern Cameroon

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Abstract

It is commonly recommended to incorporate diesel generators into distributed hybrid renewable energy systems (HRESs) to lower the system's total cost and make the generated electricity affordable. Due to the environmental and economic issues associated with fossil fuel use, biomass power technologies (BPT) appear to be an attractive alternative to diesel generators. The HOMER software was used to model, simulate, and optimise two photovoltaic/wind/battery systems integrated with different BPT (anaerobic digestion or gasification) to satisfy the electrical needs of Babadam, a remote community in northern Cameroon. The results indicated that the anaerobic digester integrated system's overall optimal architecture included a 98.1 kW photovoltaic array, a 30 kW biogas generator, and 200 batteries, with a cost of energy (COE) of \$0.347/kWh. On the other hand, the gasifier integrated option is made up of an 81.8 kW photovoltaic array, a 15 kW syngas generator, and 200 batteries and has a COE of \$0.319/kW. Additionally, compared to the PV/Wind/Battery system, integrating the biogas (resp. syngas) generator showed a potential COE decrease of 29% (resp. 40%). The sensitivity analysis highlighted the validity of this COE reduction potential everywhere in sub-Saharan Africa, leading to the conclusion that integrating BPT into HRESs can effectively contribute to Sustainable Development Goal 7.

1 | INTRODUCTION

In today's world, energy, especially electricity, is a necessity for daily life. It is the driving force behind all social, human, and economic progress [1, 2]. Electricity, on the other hand, is considered a luxury in some parts of the world. According to the International Energy Agency's World Energy Outlook, 771 million people (10% of the world population) had no access to electricity in 2019 [3]. These people live in developing countries, as seen in Figure 1 of the world map of the population without access to electricity in 2019. sub-Saharan Africa (SSA) is home to 579 million of these people, or more than two out of every three. Furthermore, 90% of the estimated 650 million people without power in 2030 will be in SSA, mostly in rural areas [4]. Therefore, rural electrification in SSA should be given special attention to achieve Sustainable Development Goal 7 (SDG 7), which explicitly aims to 'ensure access to affordable, reliable and modern energy for all.'

Grid extension and diesel generators have long been the primary modes of rural electrification in developing countries [5]. However, these modes show their limitations in addressing rural electrification challenges in the developing world [6–8]. On the other hand, various studies have highlighted distributed hybrid renewable energy systems (HRESs) as a more effective way of providing power to remote locations in developing nations, particularly in SSA [9, 10].

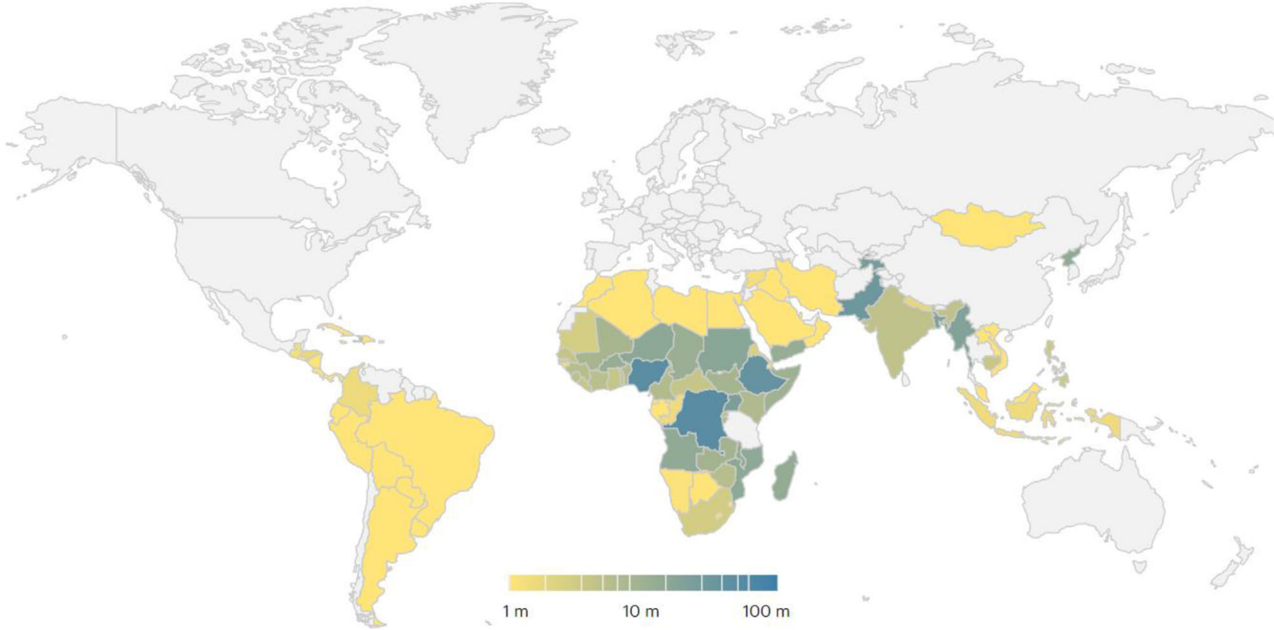

FIGURE 1 Population without electricity in 2019 [3]

A distributed system is characterised by small generation units ranging in size from a few kilowatts (kW) to 50 megawatts, usually near consumer loads or distribution and subtransmission substations [11]. An HRES is a multi-component power system fuelled by renewable and/or fossil energy sources integrated with an energy storage system and electronic devices [12]. The HRES-related literature is dominated by optimal design and techno-economic feasibility studies [13, 14]. Most of these studies indicate that the energy storage device accounts for a substantial portion of the system's overall cost [15, 16]. Using fossil fuel generators in these systems is often advocated to reduce the required size of the energy storage device, thereby lowering the overall system cost and making the generated power more affordable to impoverished communities in developing countries. Hence, PV-Wind-Diesel, PV-Diesel-Battery, Wind-Diesel-Battery, and PV-Wind-Diesel-Battery are among the most popular configurations of HRESs for rural electrification.

Nonetheless, growing concern about global warming and pollution is leading to the replacement of fossil fuel generators with greener alternatives. As such, biomass generators may be a more preferred option due to their reduced carbon footprint and lower investment and production costs [17]. Biomass is a broad term that encompasses any biological resource convertible into energy, including plant-based materials (forest and agricultural wastes) and animal-based materials (human and animal wastes, animal corpses, and living beings of the soil). While biomass has traditionally been used for cooking and heating in rural communities, it can also be used to generate electricity through two categories of technology: thermochemical and biological. Thermo-chemical processes are further classified into two types: combustion and gasification. In combustion technology, biomass is burned in a boiler with more than a stoichiometric volume of air. The power block produces electricity by using the heat produced by the boiler. In the gasification process, biomass is subjected to a sequence of chemical reactions: drying, pyrolysis, oxidation, and reduction in the presence of a gasifying agent such as air, oxygen, or steam. In biological technology, anaerobic digestion turns biomass into biogas through four stages: hydrolysis, acidogenesis, acetogenesis, and methanogenesis. Figure 2 depicts these technologies. Detailed information about biomass power technologies is available in [18, 19].

Over the past decade, there has been an increase in research into biomass-based HRESs. This is the case of the study by Ifegwu and Anjaneyulu [20]. They used Hybrid Optimisation of Multiple Energy Resources (HOMER) Pro software to optimise and analyse a PV/Biomass/Battery system to provide power and potable water to Umudike, a small community in south-eastern Nigeria. Their findings revealed that the optimal system, which comprised a 24 kW biogas generator, a 10 kW PV array and one battery string, could meet the community's electricity needs at the cost of energy (COE) of 0.118 \$/kWh. The system also supplied the community with approximately 333 litres of potable water using exhaust heat from the biomass power generator to a Scarab AB Air-Gap Membrane Distillation System. Suresh et al. [21] applied both HOMER Pro software and genetic algorithm (GA) in modelling and optimising various combinations of biomass-based HRESs to supply electricity to a cluster of three off-grid and remote villages in Karnataka state, India. They considered the system's total net present cost (NPC) as the objective function to be minimised. The least-cost configuration in both HOMER Pro and GA simulations was the PV/wind/biomass/fuel-cell combination. However, while the COE in the HOMER Pro simulation

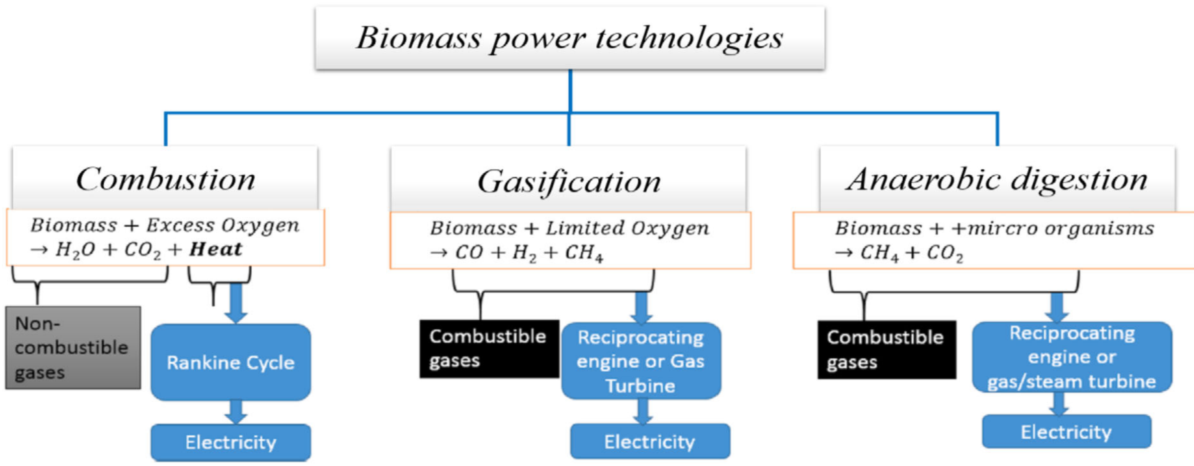

FIGURE 2 Different biomass power technologies

was 0.214 \$/kWh, it was 0.163 \$/kWh in the GA simulation. Chowdhury et al. [22] deployed RETScreen simulation software to analyse a biomass/PV/battery system for the power supply of Ashuganj, a remote area of Bangladesh. The results of their study revealed that the project of replacing a kerosene-based system with the proposed system had a simple payback of 6.9 years and a carbon dioxide savings potential of 3.81 tonnes per year. Hence, they concluded that the proposed system was environmentally friendly and less costly than the existing kerosene-based system.

Malik et al. [23] used HOMER Pro software to model three biomass-based HRESs to supply electricity to an educational building in India's western Himalayan hilly region. They found that PV/biomass/grid was the best configuration with a COE of 0.102 \$/kWh, a renewable energy fraction of 83% and a CO_2 potential savings of 27.8 Mt/year. Jameel et al. [24] applied HOMER Pro to the optimal design of a grid-connected biomass/PV/wind hybrid system for the electricity supply of Kallar Kahar, a village in Punjab province, Pakistan. Their results revealed that, for a peak load of 73.6 MW, the optimal system could generate electricity at a COE of 0.0525 \$/kWh and a renewable fraction of 87.7%. Balamurugan et al. [25] used the HOMER software to optimise a stand-alone biomass/PV/wind/battery for electrifying three selected villages in India. Their results showed that the proposed system could meet village electricity needs at a COE of 0.1095 \$/kWh. Jahangir et al. [26] performed the optimal design and sensitivity analysis of an off-grid biomass/PV/wind/battery system for Fars Province, India. Their results displayed an optimal system of a 150 kW biogas generator and an 80.7 kW PV array. The sensitivity analysis results showed that the COE varied between 0.129 and 0.223 \$/kWh according to the biomass price and inflation rate. Li et al. [27] used a multi-objective stochastic model to optimise a biomass/PV/wind hybrid system for power supply to Carabao Island in the Philippines. The results showed that the optimal system consisted of 1038 photovoltaic panels, 47 wind turbines, a 257 kW biomass generator, a 49 kW gasifier, a 77 kW battery and a 4 kW pyrolysis component. With a daily net cash flow of 455 \$/day and a CO_2 savings potential of 2795 kg/day, the system has proven to be more cost-effective and environmentally friendly than traditional electrification modes.

Murugaperumal and Raj [28] also used HOMER Pro software for optimal design and economic analysis of a biomass/PV/wind/battery system to elect Korkadu, a remote village in India. Their findings illustrated that the optimal system consisted of a 50 kW PV array, one 6 kW wind turbine, and a 100 kW biomass generator for an NPC, COE and RF of INR 2,160,000 INR; INR/kWh and 0.78, respectively. Suresh et al. [29] applied thermodynamic modelling to sizing and analysing a solar thermal/biomass system to satisfy the power needs of Rajasthan, a remote locality in India. Their analysis revealed that the hybrid system could generate electricity at a COE of 6 INR/kWh for a biomass plant capacity above 5 MW if the cost of biomass is 2500 INR/tonne. Yimen et al. [30] applied HOMER Pro to modelling and sizing of a pumped-hydro/biomass/PV/wind for rural electrification of Djoundé in northern Cameroon. The optimal system configuration included a 15 kW biogas generator and an 81.8 kW PV array, with an NPC of €370,426 and a COE of €0.256/kWh. Pavankumar et al. [31] used a multi-objective artificial cooperative search algorithm to solve the optimal design problem of a grid-integrated biomass/PV/wind/battery system for a village in Andhra Pradesh, India. Their analysis showed that introducing a biomass gasifier into the proposed HRES made the system more reliable and the COE less.

All of the studies selected above are summarised in Table 1. This summary shows that the majority of these studies took place in Asia, with most cases in India. In SSA, such applications are sparse. The leading biomass power technologies in these studies are anaerobic digestion and gasification. These technologies make use of three primary feedstocks: livestock manure, forest waste and agricultural residues. Combustion technology

TABLE 1 Selected studies on biomass-based HRES in developing countries

Authors [Ref.]	COA	System components	Technique/software	Biomass resources	BPT
Ifegwu et al. [20]	Nigeria	BG/PV/wind/BAT	HOMER Pro	Agricultural residues	Gasification
Suresh et al. [21]	India	BG/PV/wind//FC	HOMER Pro & GA	Forest residues	Gasification
Chowdhury et al. [22]	Bangladesh	BG/PV/BAT	RETSCREEN	Livestock manure	AD
Malik et al. [23]	India	BG/PV/Grid	HOMER Pro	Forest residues	Gasification
Jameel et al. [24]	Pakistan	BG/PV/Grid	HOMER Pro	Livestock manure	AD
Balamurugan et al. [25]	India	BG/PV/wind/BAT	HOMER Pro	Agricultural residues	Gasification
Jahangir et al. [26]	India	BG/PV/wind/BAT	HOMER Pro	Agricultural residues	Gasification
Li et al. [27]	Philippines	BG/PV/wind	M.O.S.M.	Forest residues	Combustion
Murugaperumal et al. [28]	India	BG/PV/wind/BAT	HOMER Pro	Livestock manure	AD
Suresh et al. [29]	India	BG/Solar thermal	TM	Forest residues	Combustion
Yimen et al. [30]	Cameroon	BG/PV/wind/PH	HOMER Pro	Livestock manure	AD
Pavankumar et al. [31]	India	BG/PV/wind/BAT	MOACS algorithm	Livestock manure	AD

Abbreviation: COA, Country of Application; BPT, Biomass Power Technology; BG, Biomass Generator; PV, Photovoltaic; BAT, Battery; FC, Fuel-Cell; GA, Genetic Algorithm; AD, Anaerobic Digestion; MOSM, Multi-Objective Stochastic Model; TM, Thermodynamic Modelling; PH, Pumped Hydro; MOACS, Multi-Objective Artificial Cooperative Search.

is less prevalent since biomass combustion, like fossil fuel combustion, generates carbon dioxide. While the carbon dioxide emitted equals the carbon dioxide absorbed from the atmosphere by trees and plants during photosynthesis, these releases might be avoided by applying anaerobic digestion or gasification technologies instead. Apart from CO2, biomass fuel combustion releases various other toxic gases into the atmosphere, including carbon monoxide, NOx (nitrogen oxides), and VOCs (volatile organic compounds), the majority of which lead to air pollution.

However, to the best of the authors' knowledge, no study in the literature has performed a separate analysis of the two leading technologies fuelled by different feedstocks. Such a study would help indicate the most effective biomass power technology and the appropriate biomass resource in the study area to be incorporated into the proposed HRES. Moreover, HOMER tends to be the most often applied analysis tool in the papers reviewed. However, none of the previous studies have used the sensitivity analysis capability of the software to analyse the role that biomass generators can play in different locations with varying levels of abundance of renewable energy resources (wind speed and solar radiation).

Furthermore, the investment cost of renewable energy systems, especially solar photovoltaic, is expected to decrease significantly. The analysis of the role of biomass-based power technologies in the sustainability of future HRES with a reduced capital cost of components has not yet been carried out in the literature. Finally, biomass resources are not available everywhere, leading to an increased price of these resources due to high transportation costs. The relevance of acquiring these resources far from the implementation site of the biomass-based HRES project has never been elucidated.

This research aims to address the gaps in the literature mentioned above by using the simulation, optimisation and sensitivity analysis capabilities of HOMER software to investigate two options for distributed biomass-based HRES designed to meet the domestic, community, agricultural, and commercial loads in Babadam, a remote location in northern Cameroon.

The rest of this paper is organised as follows: Section 2 describes the materials and methods used to conduct the analysis. Section 3 summarises the findings and discusses them, while Section 4 concludes the paper.

2 | MATERIALS AND METHOD

Two approaches to evaluating and optimising HRES have been described in the literature: software tools and optimisation methods. Luna-Rubio et al. [32] and Connolly et al. [13] have provided an in-depth examination of these two categories of technique. Among software tools, the HOMER software, designed in 1992 at the National Renewable Energy Laboratory in the United States, emerges to be the most widely used, based on the literature review conducted above. It is capable of simulating, optimising, and performing sensitivity analyses. Moreover, its library has various technologies and components that enable it to be used for HRES modelling. These benefits justify the use of HOMER Pro Version 3.14.4, the most recent release, in this research. Additional information about HOMER software is available in [33]. The analysis was performed on a 64-bit version of Windows 10 Pro equipped with a 2 GHz Intel Core i7 processor, 8 GB of RAM, and a 3 GB graphics card.

Figure 3 roughly depicts the study's research methodology. A pre-HOMER process was carried out along with the HOMER analysis, which included a thorough investigation of the village's load, available resources, and site layout. Data from surveys, literature reviews, and expert opinions were analysed to provide HOMER-specific data throughout this process.

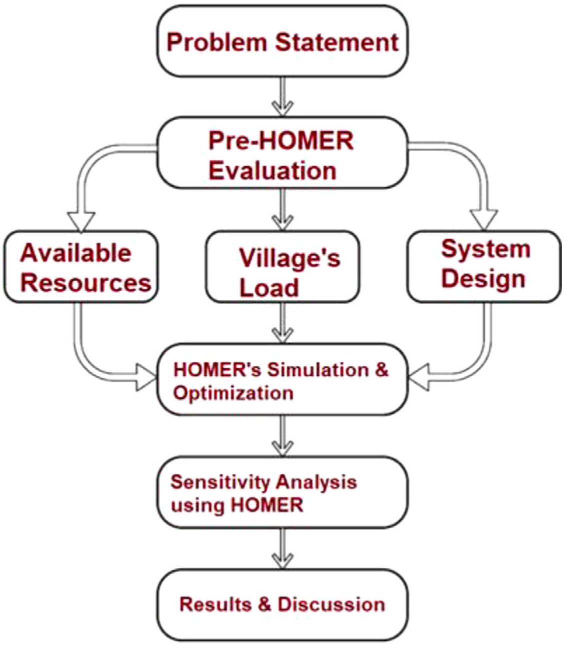

FIGURE 3 Schematic diagram of the methodology

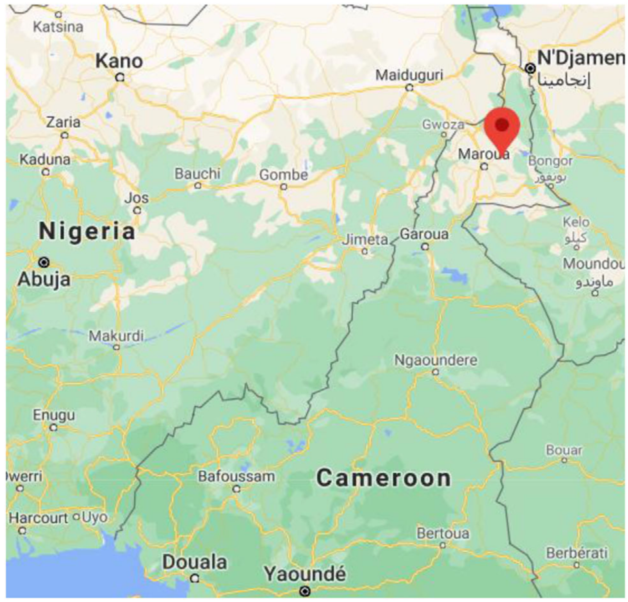

FIGURE 4 The spatial location of Babadam in northern Cameroon.

TABLE 2 Background information of the study area

Particulars	Details
Country	Cameroon
Region	Far North
Division	Diamaré
Name of the municipality	Bogo
Latitude	10° 46′ 54′′ North
Longitude	14° 14′ 16′′ East
Elevation above sea level	314 m
Population	600
Number of households	100
closest power transformer	Bogo, 31 km
Socio-economic activities	Agriculture and crafts.

The following subsections provide a detailed background on the adopted methodology.

2.1 | Study location and load assessment

The planned biomass-integrated HRES was dedicated to serving the load needs of Babadam, a rural community in Bogo district in the Far North Region of Cameroon. The community is not yet linked to the national electrical grid. The closest power transformer is in Bogo, a town located 31 kilometres away. Figure 4 presents the research area's spatial position, while Table 2 provides its background information.

Electricity is required for an adequate standard of living in most rural communities in developing countries, where it is predominantly used for lighting, motor applications, refrigeration and communication. The study area's demand load was determined for residential (100 households), commercial (two shops, one mini diary and one four-mil), community (one health centre, one primary school and street lights) and agricultural sectors. The assessment considered the location's potential needs and previous studies conducted in Nigeria, Ghana and Pakistan [12, 34, 35]. Two types of load were considered: primary load and deferrable load. A deferrable load is an electrical load that needs a certain amount of energy within a specified period but does not require precise timing; it can wait until electricity is available.

In contrast, a primary load must be satisfied within a given period (e.g. lighting systems). This form of the load is inflexible in terms of time and disruption. Appendix B details the number, energy rating and operation hours of energy-consuming appliances needed for the sectors considered for primary loads. Figure 5 shows the study area's hourly electricity demand per sector. Random variabilities of 20% day-to-day and 15% time-step were considered to provide a more realistic load profile. Figure 6 illustrates the resulting annual load profile. The average daily load, average power, peak power and load factor were 286.9 kWh, 11.95 kW, 55.88 kW and 0.21, respectively.

The deferrable load was dedicated to irrigation and domestic water pumping. In SSA, pumped water requirements are often higher during the dry season than during the rainy season since precipitation supplies a part of the rainy season's water demand. Figure 7 represents the monthly average daily deferrable load in Babadam. The average monthly daily average load was evaluated at 75 kWh/day during the dry season (May to September) and 40 kWh/day during the rainy season (October to April).

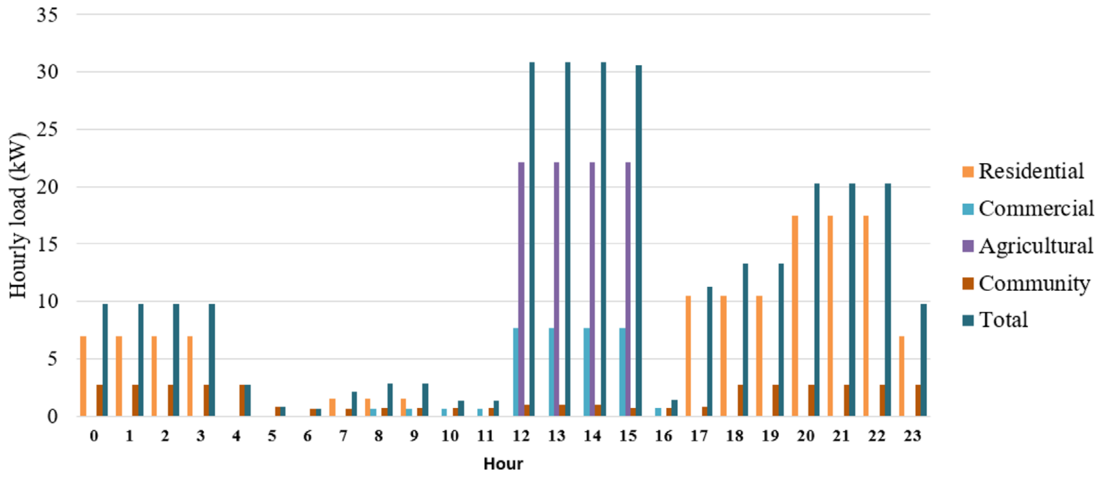

FIGURE 5 Hourly electricity load by sector

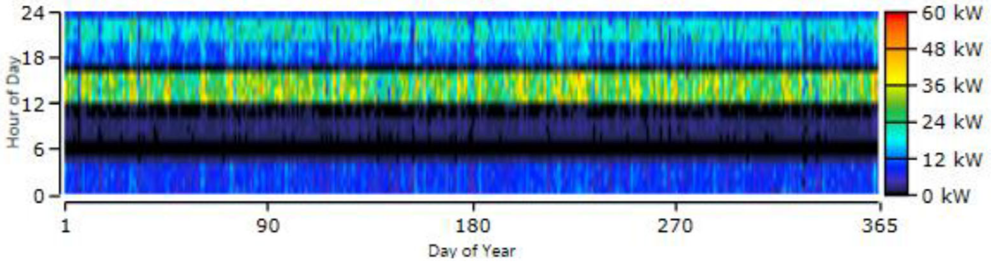

FIGURE 6 Annual load profile

Besides, the storage capacity, peak load and minimum load ratio considered in this analysis were 100 kWh, 20 kW and 40%, respectively.

2.2 | Evaluation of available energy resources

2.2.1 | Available solar and wind resources

Data on solar and wind resources for Babadam, the chosen site, were acquired from the NASA surface meteorology and solar energy (SSE) database [36] at Maroua (10°35'50" N, 14°18'57" E), the nearest data point. Figure 8 depicts the monthly average daily solar radiation (MADSR) and clearness index at that location. On a yearly average basis, solar radiation was found to be 5.71 kWh/m^2/day, and the average clearness index was 0.6. March received the highest solar radiation, 6.56 kWh/m^2/day, while August received the lowest, 4.85 kWh/m^2/day. This solar radiation data highlights the research area's significant solar energy potential for photovoltaic array-based power generation. Figure 9 illustrates monthly data on average wind speed. At the site's 50 m anemometer, the annual average wind speed was 5.14 m/s.

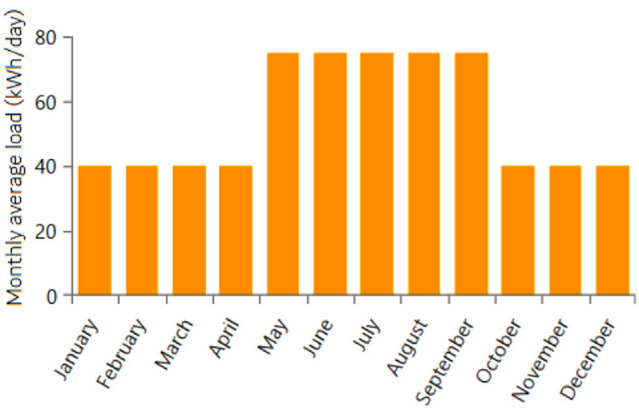

FIGURE 7 The monthly average daily deferrable load

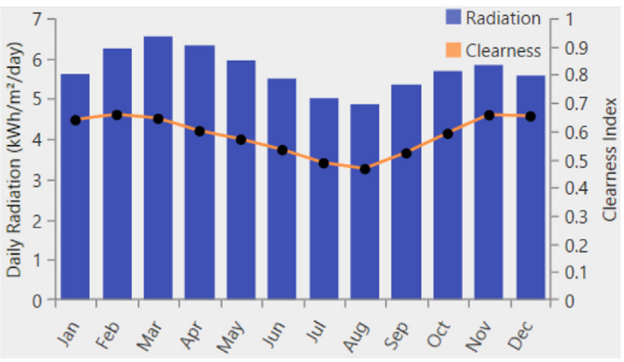

FIGURE 8 Monthly daily average solar radiation and clearness index

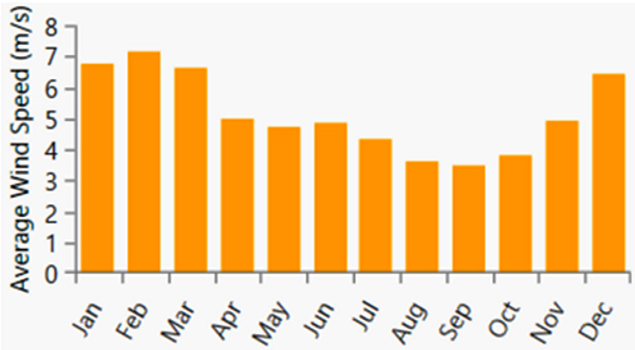

FIGURE 9 Monthly average wind speed.

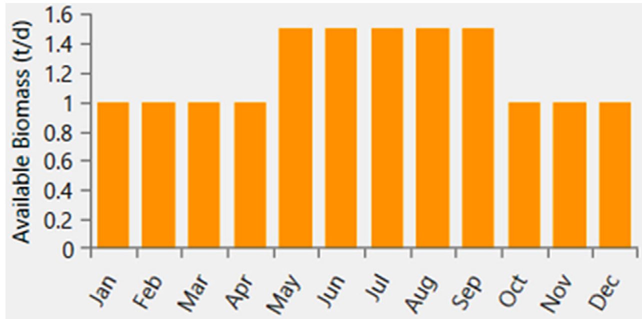

FIGURE 10 Monthly available daily forest waste

TABLE 3 Evaluation of dung availability

Animal	Number	Manure available (kg/head/day)	Total daily dung (kg/day)
Cows	1000	10	10,000
Horse	100	10	1000
Mule	200	10	2000
Goat	2500	1	2500
Total	3800		15,500
Total recoverable dung (RF=0.70)			10,850

Abbreviation: RF, recovery factor.

2.2.2 | Biomass resources

Forest residues and animal dung were the biomass resources considered for electricity production in this study. The input data of HOMER Pro was the monthly average daily supply for biogas (resp. syngas) processing, expressed in tons/day. A survey undertaken in the research area determined that the overall animal population was 3,800, comprising 1000 cows, 100 horses, 200 mules, and 2500 goats. Table 3 provides a comprehensive assessment of on-site animal dung availability. This assessment assumed that each cow/mule/horse produced ten kilograms of manure per day and that each goat produced one kilogram per day [37]. Given the implausibility of collecting all livestock manure, a recovery factor of 0.7 was considered, as was done in [30]. Therefore, daily animal manure availability was determined to be 10,850 kg/day.

Unlike available livestock manure, which is considered constant throughout the year, available forest waste varies by season. Generally, forest waste is more prevalent during the dry season than it is during the wet season. Monthly daily forest waste, as determined by a site survey, is depicted in Figure 10. Livestock manure was turned into biogas through the anaerobic digestion process, while forest waste was transformed into syngas through the gasification process. Besides the monthly average daily biomass resource, HOMER Pro also requires data on the average price of biomass (t/$), carbon content (%), gasification ratio (kg/kg) and the lower heating value of the biogas (MJ/kg). The associated values of these parameters were determined and reported in Table 4 for each resource based on previous studies [38–41].

TABLE 4 biomass resource input in HOMER

Biomass resource	Average price (t/$)	Carbon content (%)	GR (kg/kg)	LHV (MJ/kg)
Livestock manure	15	35	0.03	20
Forest waste	20	45	2.4	10

Abbreviation: GR, gasification ratio; LHV, lower heating value.

2.3 | System configuration and component modelling

Figure 11 shows the schematic design of the two HRESs being examined. The two systems include three generators each (a photovoltaic array, a biogas or syngas generator and a wind turbine), an energy storage unit, a control station, a conversion system and a load. In the biogas-integrated system, the biogas generated by an anaerobic digester from livestock manure feeds the biogas power generator (BPG). In contrast, in the gasifier-integrated design, the gasifier provides syngas from forest waste to the syngas power generator (SPG). The battery bank stores excess electricity generated by intermittent renewable energy sources (PV array and wind). The biogas (resp. syngas) generator serves as a backup power source triggered whenever the combined output from the photovoltaic array, wind turbine, and battery is inadequate.

2.3.1 | PV array

Sunpower's SPR-X20-327 monocrystalline module was selected as the photovoltaic solar cell in this study. The module has a maximum output voltage of 600 V DC and a power rating of 327 Wp. Table 5 [42] summarises the technical details for this photovoltaic module. To estimate the output of a photovoltaic array, HOMER uses the following equation[43]:

$$P_{output} = Y_{PV}\ f_{PV} \left(\frac{G_T}{G_{T,STC}} \right) \left[1 + \alpha_P \left(T_c - T_{c,STC} \right) \right] \quad (1)$$

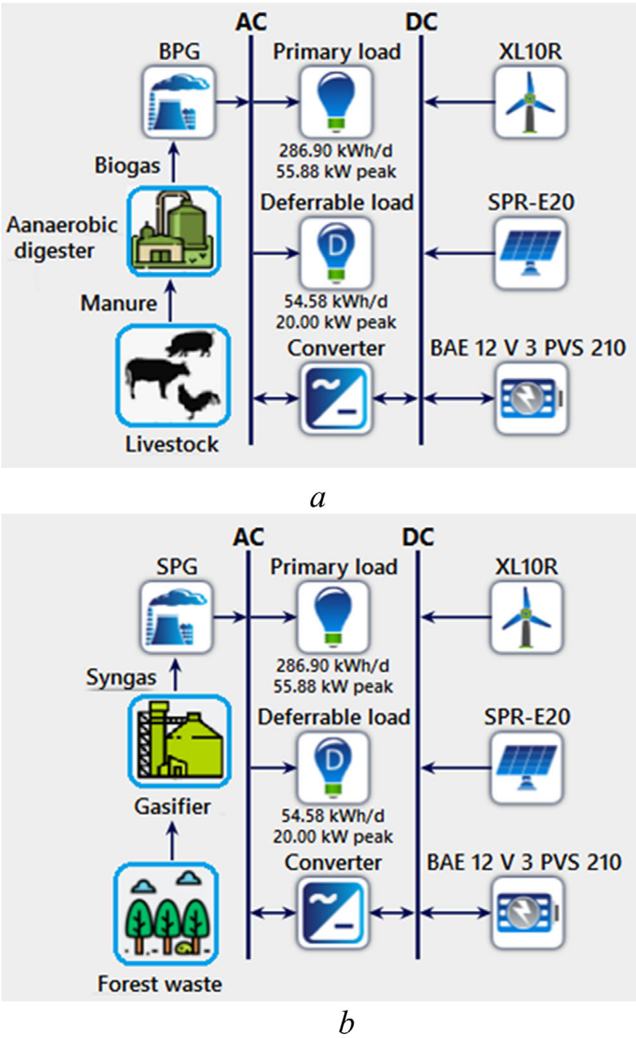

FIGURE 11 Biomass-based HRE options. (a) Anaerobic digester integrated option, (b) Gasifier-integrated option

where Y_{PV} (kW) and f_{PV} (%) represent the photovoltaic's power rating and derating factor, respectively. G_T (kW/m^2) and $G_{T,\,STC}$ (= 1 kW/m^2) denote the global and standard (at 25°C) solar radiation incident. T_c (°C) and $T_{c,\,STC}$ (°C) indicate the photovoltaic cell temperature undercurrent and standard conditions, respectively. α_p (%/°C) means the power temperature coefficient. The Graham and Holland algorithm [44] was applied to generate HOMER software's hourly global solar radiation from the monthly average global solar radiation. The life span of photovoltaic panels was predicted to be 25 years. Investment costs, replacement costs, and operational costs were estimated at \$3,000/kW, \$3,000/kW, and \$10/kW/year, respectively [45].

2.3.2 | Wind turbine

In this research, Bergey Windpower's excel 10-R wind turbine was considered. Its nominal output is 10 kW at a wind speed of 12 m/s. Table 6 [46] provides the specifications of the turbine.

TABLE 5 Characteristics of the selected PV module

Item	Specification
Manufacturer	Sunpower
Module type	Monocrystalline
Module model	SPR-X20-327-COM
Power rating	327 W
Module peak efficiency	20.5%
Open circuit voltage (VoC)	64.9 V
Rated voltage (Vmpp)	54.7 V
Power tolerances	–3%/+2%
Maximum voltage	DC 600 V
Rated current (Impp)	5.98 A
Current temp coef	3.5mA/°C
Short circuit current (ISC)	6.46 A
Temp. coef. of voltage	–175 mV/°C
Temp. coef. of power	–0.38%/°C
Operating temperature	–40°C to +85°C
Weight	18.6 kg
Power per unit area	0.2005 kW/m^2

TABLE 6 Technical characteristics of excel 10-R wind turbine

Item	Specification
Nominal power	10 kW at 12 m/s
Wind turbine model	Bergey excel 10-R
Manufacturer	Bergey Wind Power
Operating temperature	–40°C to +60°C
Cut-in wind speed	2.5 m/s
Rotor diameter	7 m
Max. design wind speed	60 m/s
Cut-Out wind speed	None
Furling wind speed	14 m/s–20 m/s
Net weight	545 kg
Hub height	30 m
Type	3 Blade Upwind

The power law was used to compute wind velocity at the hub height:

$$V_{hub} = V_{anem} \cdot \left(\frac{H_{hub}}{H_{anem}} \right)^{\alpha} \tag{2}$$

where V_{anem} and V_{hub} denote wind speeds at the anemometer and hub heights (H_{anem} and H_{hub}), respectively. α indicates the exponent of the power-law equation, which is set at 0.14. Capital, replacement, and operations and maintenance costs for wind turbines were established at \$50,000/unit, \$30,000/unit, and \$200/year, respectively [45]. Additionally, the turbine was expected to have a 20-year life.

TABLE 7 Economic and technical characteristics of the battery model adopted

Item	Specification
Manufacturer	BAE
Model	BAE PVV Block 12V 210
Nominal capacity	2.42 kWh
Nominal voltage	12 V
Capacity ratio	0.375
Maximum capacity	201 Ah
Battery charge efficiency	90%
Rate constant	1.06 (1/h)
Maximum charge current	33.4 A
Battery discharge efficiency	100%
Maximum charge rate	1A/Ah
Maximum discharge current	361 A
Hourly self-breadischarge rate	0.0002
Maximum DOD	70%
Lifetime	5-year life
Capital cost	$200/unit
Replacement cost	$200/unit
O&M cost	$0/unit/year

2.3.3 | Biomass generators

Both options were modelled using a generic biomass generator connected to an alternating current output. When sizing the generator, HOMER takes the available resources into account. The authors took the investment and replacement cost per kW at 2000 and 1500 $/kW for the BPG and 6000 and 5500 $/kW for the syngas generator. The maintenance cost for both types of generators was set at $0.1/h and their life expectancy at 15,000 h [47]. The minimum load ratio was estimated at 30% of the rated power.

2.3.4 | Battery bank

The most often used energy storage device in HRESs is the battery, specifically the deep-cycle lead-acid battery. A thorough examination of the battery's charge and discharge specifications is critical for deciding the battery's optimum number. In this work, we considered the BAE PVV Block 12V 210 model of battery. This model has a maximum depth of discharge (DOD) of 70%. The economic and technical characteristics of this model are summarised in Table 7.

2.3.5 | Converter

A power converter system is required to maintain power flow between the system's alternating current (AC) and direct current (DC) components. This study considered a generic system converter composed of a rectifier and an inverter for accomplishing bidirectional AC-DC conversion. The costs of investment, replacement and operations and maintenance were considered to be $650/kW, $600/kW, and $0/kW [38], respectively. The efficiency of the inverter and rectifier was adjusted to 95%. The converter was projected to have a 15-year life.

2.4 | System simulation and optimal design

2.4.1 | Performance criterion

The NPC is the assessment criterion for HOMER software. Based on NPC, HOMER rates all feasible combinations and determines the optimal system. It is computed as the total of all discounted system expenses and revenues. Capital costs, replacement costs, operating costs and fuel costs were included in the expenses, while revenues were the system component salvage values. The NPC of an HRES is calculated as follows [48]:

$$NPC = \frac{C_{ann,tot}}{CRF\ (d,\ N)} \tag{3}$$

where $C_{ann,tot}$ denotes the total annualised cost of the system, CRF is the capital recovery factor, N is the project lifetime, and d denotes the discount rate. The total annualised cost of the system is expressed as:

$$C_{ann,tot} = C_{ann,cap} + C_{ann,rep} + C_{ann,O\&M} + C_{ann,fuel} - R_{ann,salv}, \tag{4}$$

where $C_{ann,cap}$ is the annualised capital costs, $C_{ann,rep}$ is the annualised replacement costs, $C_{ann,O\&M}$ is the annual operations and maintenance costs, $C_{ann,fuel}$ is the annualised fuel costs, and $R_{ann,salv}$ is the total annualised savage values of all system components. The (CRF) converts a present value to a uniform yearly cash flow series over the project's life (N) at a specified discount rate (d). The CRF formula is as follows:

$$CRF = \frac{d(1+d)^N}{(1+d)^N - 1} \tag{5}$$

The project's lifespan was estimated to be 20 years in this analysis. The calculation of the discount rate is based on the following formula:

$$d = \frac{i - f}{1 + f} \tag{6}$$

where f denotes the annual rate of inflation, and i denotes the nominal interest rate. Annual inflation and nominal interest rates were set at 3% and 8%, respectively. HOMER determines

TABLE 8 Optimisation variables and search space of the proposed model

Optimisation variable	Number of wind turbines	PV array size (kW)	Number of battery strings	Size of biomass generator (kW)	Converter capacity (kW)
Maximum	9	163.5	150	35	100
Minimum	0	0	0	0	0
Step	1	16.35	10	5	10

TABLE 9 Sensitivity variables and corresponding values

Sensitivity variable	Values
Solar radiation (kWh/m^2/d)	3.8, 5.7, 7.8
Wind speed (m/s)	3, 5.08, 7, 9
Solar PV cost multiplier ($/kW)	0.4, 0.6, 0.8, 1
Forest waste price ($/t)	20, 30, 40, 50, 60
Livestock manure price ($/t)	10, 20, 30, 40, 50

the COE in addition to the NPC. This is the total cost of generating one kilowatt-hour of electricity by the system. It is defined as the system's total annualised expense ratio to its total annual usable electricity output. The COE's formula is as follows:

$$COE = \frac{C_{ann,tot} - C_{boiler} \cdot H_{thermal}}{E_{served}} \quad (7)$$

where $C_{ann,tot}$ denotes the system's total annualised cost (in dollars per year), C_{boiler} is the boiler's marginal cost (in dollars per kWh), H_{served} is the total thermal load served (in kWh per year), and E_{served}, the total electrical load served (in kWh per year). Thermal load is not served in this analysis, thus $H_{served} = 0$.

2.4.2 | Dispatch strategy

A dispatch strategy is a system of principles that regulates the operations of the generator(s) and storage unit(s). The HOMER software can dispatch in two modes: load following (LF) and cycle charging (CC). When the load-following approach drives a generator, it generates just the power required to fulfil demand. In contrast, when a generator is activated under a CC strategy, it runs at full capacity, with any excess energy stored in the power storage unit. This research considered both LF and CC strategies.

2.4.3 | Optimisation variable and search space

A decision variable, alternatively called an optimisation variable, is a variable that the system developer may control. During the optimisation process, HOMER evaluates all possible combinations of optimisation variables. The optimisation variables considered in this research are summarised in Table 8, along with their corresponding values. They comprise the solar array's size (11 values), the wind turbine number (10 values), the battery string number (16 values), the biogas (or syngas) power generator size (8 values), and bidirectional converter capacity (11 values). The search space is a collection of all conceivable system designs, among which HOMER determines the optimal configuration. The optimal design should have the lowest value of the assessment criterion and be feasible, i.e. it should satisfy all constraints of the optimisation problem. Both dispatch strategies, LF and CC, were simulated for all combinations of optimisation variables. HOMER thus simulated $10 \times 11 \times 16 \times 8 \times 11 \times 2 =$ 309,760 configurations.

2.4.4 | Constraints

Constraints are mandatory requirements for HRES setups. Following the optimisation operations, a configuration that violates one of the stated constraints is considered unfeasible and therefore not evaluated by HOMER. The proposed case study considers two distinct types of constraints: first, capacity shortage constraints are defined in terms of the overall yearly capacity shortfall, set at 1% for this study. This implies that HOMER discarded any system configuration that did not meet 99% of the annual power load and the operational reserve. Second, operating reserve constraints add excess operational capacity to ensure system resilience in an unexpected increase in load or a fall in renewable energy supply. HOMER calculates the needed operational reserve using four inputs: two represented as a percentage of the fluctuation of power demand (current period step load and yearly peak load), and two expressed as a percentage of renewable energy generation (solar PV output and wind power output). The operational reserve percentages for current time step load, yearly peak load, solar PV output, and wind power output were set at 10%, 15%, 20%, and 50%, respectively, in this research study.

2.5 | Grid extension

As previously stated, grid extension is a common means of electrifying rural areas in SSA. This research considered it as a viable alternative to the proposed off-grid hybrid system. The HOMER compares the two options by calculating the break-even grid extension distance (BGED), the distance from the grid at which the proposed off-grid system's NPC is similar to that of the grid expansion. Further than this

TABLE 10 Optimisation results for the anaerobic digester integrated option

	Specification	Unit	Best hybrid system per category 1	2	3	4	5
System architecture	Solar PV array	kW	98.1	81.8	981.	131	0
	Wind turbine	Number	0	1	1	0	7
	Biogas power	kW	30	30	0	0	30
	Battery	Number	200	150	550	500	250
	Converter	kW	50	50	60	60	60
	Dispatch strategy	LF or CC	LF	LF	CC	CC	CC
Cost	COE	$/kWh	0.347	0.350	0.447	0.456	0.604
	NPC	$	546,009	550,346	703,062	716,135	950,270
	Total O & M cost	$/year	7,990	6,819	22,772	16,625	14,641
	Total capital cost	$	426,800	417,750	493,300	531,400	499,000
Power production	PV array	kWh/year	184,029	153,358	184,029	245,373	0
	Wind turbine	kWh/year	0	24,603	24,603	0	172,219
	BPG	kWh/year	4,610	6,642	0	0	31,353
	TEP	kWh/year	188,639	184,,602	208,632	245,373	203,571
	EPP	kWh/year	48,127	46,009	69,069	104,401	65,418
Capacity factor	Solar PV array	%	21.4	21.4	21.4	21.4	0
	Wind turbine	%	0	28.1	28.1	0	28.1
	Biogas generator	%	1.75	2.53	0	0	11.9
Biomass summary	TFC	tons/year	342	492	0	0	2,196
	AFD	kg/day	936	1350	0	0	6,020

Abbreviations: TEP, total electrical production; EPP, excess power production; TFC, total feedstock consumed; AFD, average feedstock per day.

distance, the planned off-grid option is recommended; grid extension is the preferable alternative closer to the grid. The capital cost per kilometre, the yearly operational cost per kilometre, and the grid electricity price are the required input parameters for HOMER's BGED calculation. This analysis estimates the grid extension's construction cost per kilometre and operations and maintenance cost per kilometre at $14.000/km and $300/year/km, respectively. In Cameroon, the average cost of grid electricity is $0.122 per kilowatt-hour.

2.6 | Sensitivity analysis

Sensitivity analysis examines the influence of changes in certain parameters on the optimal system. The term "parameter" refers to any numerical data that is entered into HOMER. Sensitivity variables are often used to refer to sensitivity analysis parameters. The designer enters a set of values (sensitivity values) for each sensitivity variable into HOMER. Wind turbines and photovoltaic arrays are the primary energy generators in the systems under analysis. The biogas or syngas generator is only used as a backup engine if the total output is inadequate. So, solar radiation and wind speed can affect the optimal configuration and the share of biomass power produced by the system. These two resources vary considerably throughout SSA. Solar radiation ranges from 3.8 kWh/m^2/day in Equatorial Guinea, Gabon, southern Cameroon and Congo to 7 kWh/day in Eritrea, northern Chad, northern Sudan and northern Mali [43]. The average wind speed may surpass 8 m/s in some parts of the South and East African coasts, while it is less than 4 m/s in Congo, southern Cameroon, Liberia, Gabon and Equatorial Guinea [44]. Carrying out the sensitivity analysis of the impact of solar radiation and wind speed can help provide insight into the role that biomass power generators can play in the sustainability of HRESs throughout SSA. Besides, biomass resources may not be available in some rural areas of SSA, and increased transportation costs may result in an overall cost increase. Performing a sensitivity analysis of the effect of the feedstock cost on the optimal system may inform us about the relevance of acquiring these resources far from the study area. The sensitivity analysis of the effect of the solar PV cost decrease was also carried out to get insight into the future biomass-based HRESs considering the upcoming cost decrease of solar PV.

Consequently, solar radiation, biomass price, wind speed, and solar PV investment cost were all considered sensitivity variables in this analysis. Table 9 lists all of these sensitivity variables and their corresponding values.

3 | RESULTS AND DISCUSSION

3.1 | Optimal design results

The calculation reports indicated that of the 309,760 possible configurations simulated by HOMER software for each option of biomass-based HRES, only 118,780 were feasible for the biogas-integrated system while 131,951 were for the syngas-integrated system.

Feasible configurations for the biogas-integrated option were classified into five categories of system architecture, namely: category 1 (PV/BPG/Battery), category 2 (PV/Wind/BPG/Battery), category 3 (PV/Wind/Battery), category 4 (PV/Battery), and category 5 (Wind/BPG/Battery). In contrast, feasible configurations for the syngas-integrated option were classified into seven categories: category 1 (PV/SPG/Battery), category 2 (PV/Wind/SPG/Battery), category 3 (Wind/SPG/Battery), category 4 (PV/Wind/Battery), and category 5 (PV/Battery), category 6 (SPG/Battery), and category 7 (PV/Wind/SPG). The techno-economic characteristics of the best configuration in each category are listed in Tables 10 and 11 for the biogas-integrated and syngas-integrated options, respectively.

For both options, the overall optimal hybrid system is that of Category 1. For the biogas-integrated option, the optimal system included a 98.1 kW photovoltaic array, a 30 kW biogas generator, 200 batteries, and a 50 kW bi-directional converter, and a load-following dispatch strategy. It had an NPC of \$546,009 and a COE of \$0.347/kWh. The best configuration for the syngas-integrated option comprised an 81.8 kW photovoltaic array, a 15 kW syngas generator, 200 batteries, and a 50 kW bi-directional converter with a cycle-charging dispatch strategy. It had an NPC of \$501,069 and an electricity cost per kilowatt-hour (kWh) of \$0.319. Neither of the overall optimal configurations included wind turbines.

Figure 12 depicts the discounted cash flow by component of the optimal system for each option considered. The system's overall capital cost is the most significant cost type for both options. It accounted for 78% and 80% of the NPC, respectively, for the biogas and syngas integrated options. The total capital cost for both options was dominated by the photovoltaic array, which accounted for 69% and 61% of the total capital cost for biogas and syngas integrated options, respectively. The batteries and system converter were replaced with the biogas-integrated option throughout the project's length, totalling \$90,838 in discounted replacement costs. In addition to these components, the syngas generator was also replaced in the syngas-integrated option at a total discounted replacement cost of \$117,387.

The monthly output of electricity production for both options is presented in Figure 13. In both scenarios, photovoltaic panel production forms a significant part of annual electricity production (97% and 93% for biogas and syngas integrated options, respectively). For both options, biomass

TABLE 11 Optimisation results for the gasifier-integrated option

	Specification	Unit	Best hybrid system per category						
			1	2	3	4	5	6	7
System architecture	Solar PV array	kW	81.8	81.8	0	98.1	131	0	147
	Wind turbine	Number	0	1	4	1	0	0	6
	syngas power	kW	15	15	30	0	0	35	35
	Battery	Number	200	150	100	550	500	100	0
	Converter	kW	50	40	40	60	60	30	40
	Dispatch strategy	LF or CC	CC	CC	CC	CC	CC	CC	LF
Cost	COE	\$/kWh	0.319	0.331	0.444	0.447	0.456	0.546	1.06
	NPC	\$	501,069	519,789	698,325	703,062	716,135	859,367	1.66×10^6
	Total O & M cost	\$/year	7,990	6,819	22,772	16,625	14,641	49,722	555.511
	Total capital cost	\$	400,250	433,750	411,000	493,300	531,400	232,000	959;950
Power production	PV array	kWh/year	153,358	153,358	0	184,029	245,373	0	276,044
	Wind turbine	kWh/year	0	24,603	98,411	24,603	0	0	147,616
	SPG	kWh/year	12,848	12,,157	62,004	0	0	136,804	68,832
	TEP	kWh/year	166,206	177,961	160,415	208,632	245,373	136,804	492,492
	EPP	kWh/year	26,094	51,583	24,642	69,060	104,401	0	364,486
Capacity factor	Solar PV array	%	21.4	21.4	0	21.4	21.4	0	21.4
	Wind turbine	%	0	28.1	28.1	28.1	0	0	28.1
	Syngas generator	%	9.78	9.25	23.6	0	0	44.6	22.5
Biomass summary	TFC	tons/year	11.3	10.7	54.3	0	0	120	64.2
	AFD	kg/day	30.95	29.31	148.76	0	0	328.76	175.89

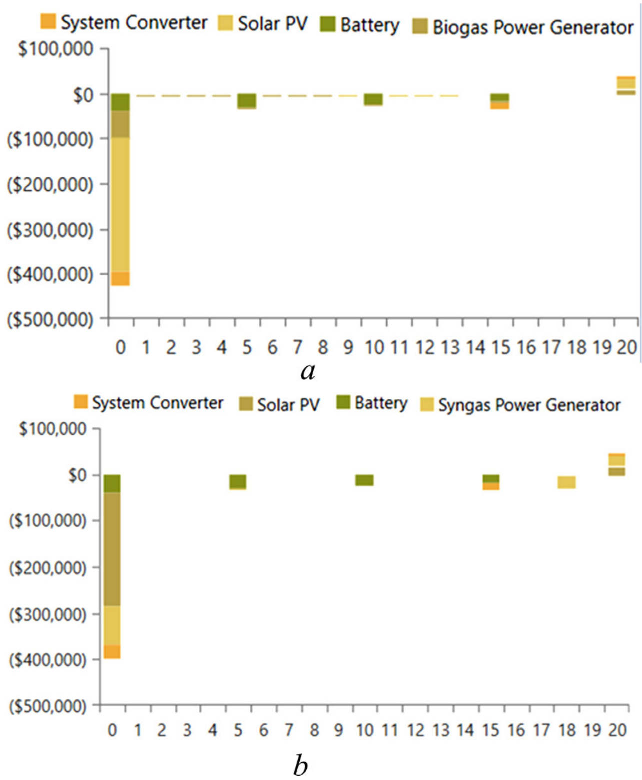

FIGURE 12 Discounted cash flow by component. (a) Anaerobic digester integrated option, (b) Gasifier-integrated option

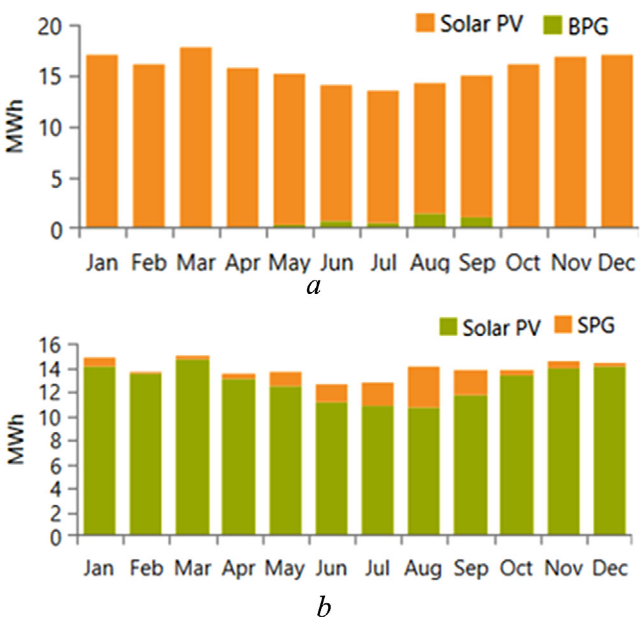

FIGURE 13 Monthly output of electricity production. (a) Anaerobic digester integrated option, (b) Gasifier-integrated option

electricity generation is more abundant during the dry season (May to September), when the deferrable load is higher.

The excess power and unmet load for the biogas-integrated option were 48,127 and 0 kWh/year, respectively, or 25.5 and 0% of the total output. On the other hand, the excess power and unmet load for the syngas-integrated alternative were

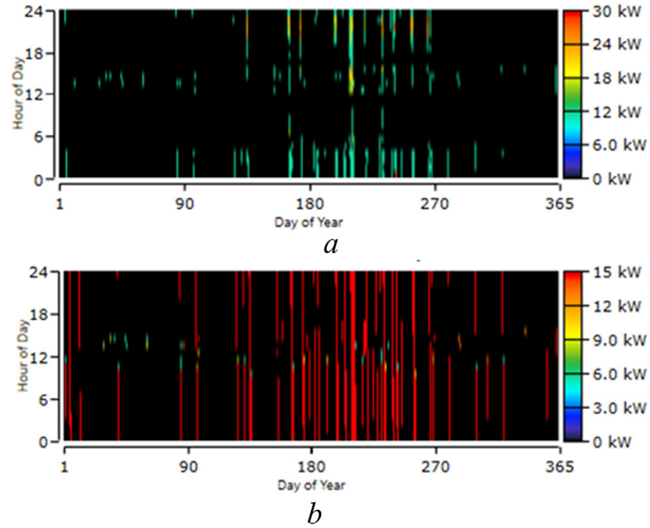

FIGURE 14 Biomass generator output profile. (a) Anaerobic digester integrated option, (b) Gasifier-integrated option

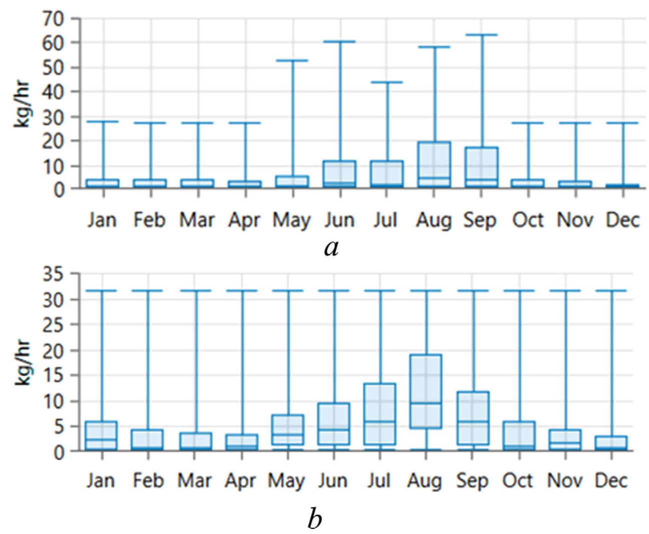

FIGURE 15 Average hourly monthly fuel consumption. (a) Anaerobic digester integrated option, (b) Gasifier-integrated option

26,094 kWh/year (16%) and 75 kWh/year (0.06%). Excess electricity is idled and released once the electrical load demand is met and the batteries are completely charged. In comparison, an unmet load cannot be fulfilled because of a mismatch between total electrical demand and total electrical generation capability. As a result, the capacity shortage of the system was 35.4 kWh/year or 0.03 % of the demand load for the biogas-integrated option, and 105 kWh/year (0.08%) for the syngas-integrated option.

The biomass generator output profile for the two alternatives is represented in Figure 14, and the related average hourly monthly fuel consumption in Figure 15. For both systems, the highest average hourly feedstock consumption occurs in August. The yearly total feedstock consumption was 342 tons for the biogas-integrated option and 11.3 tons for the syngas integrated alternative.

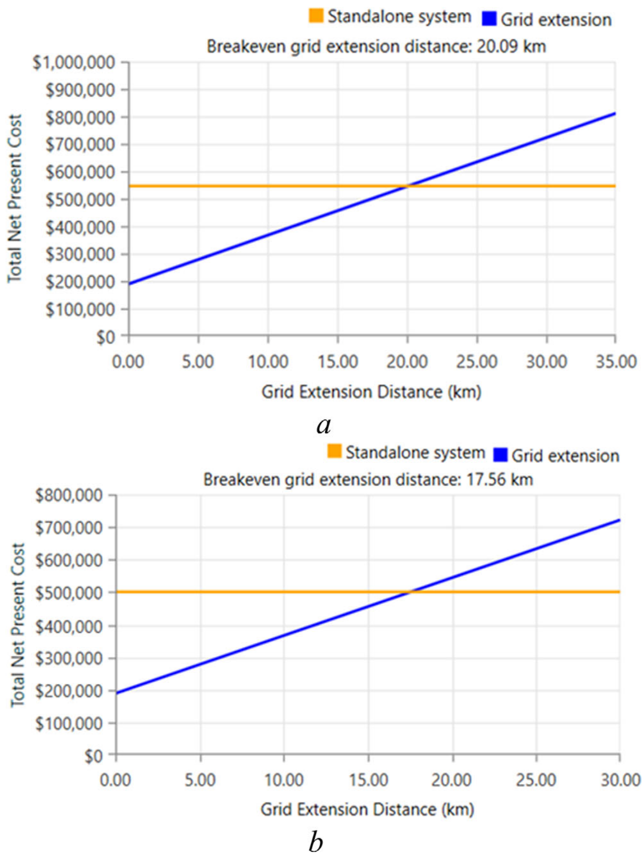

FIGURE 16 Break-even grid extension distances of the optimised biomass-based HRES. (a) Anaerobic digester integrated configuration, (b) Gasifier-integrated configuration

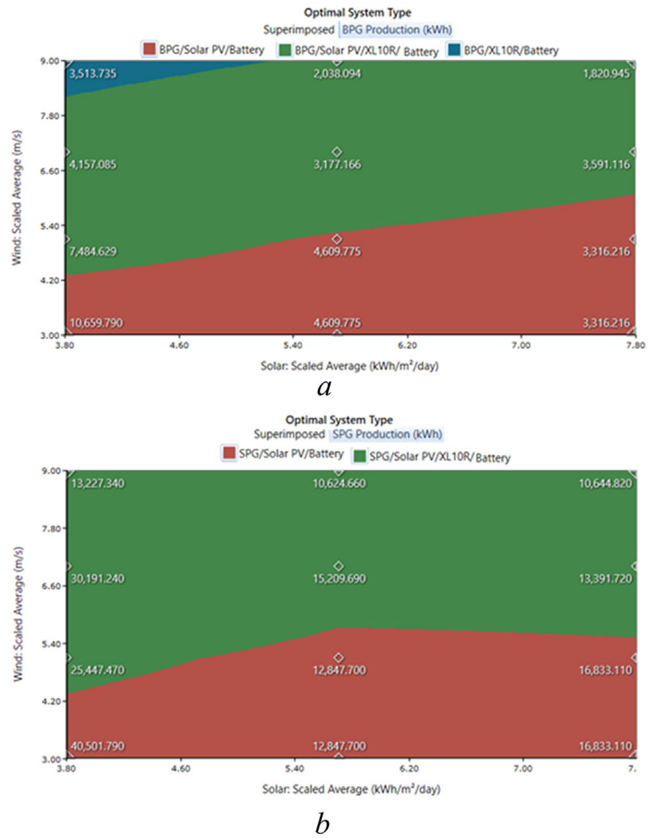

FIGURE 17 Sensitivity analysis results of wind speed and solar radiation. (a) Anaerobic digester integrated option, (b) Gasifier-integrated option

A comparison of the NPCs of the proposed autonomous systems and the grid extension for electrifying Babadam is depicted in Figure 16. This means (BGED) of 20.9 km and 17.56 km for the biogas and syngas integrated options, respectively. Consequently, both proposed alternatives are preferable to the grid extension for electrifying Babadam, as the closest power transformer is situated in Bogo, 31 km away.

3.2 | Sensitivity analysis results

Figure 17 depicts the sensitivity analysis results of solar radiation and wind speed on the optimal system. Within the spectrum of these two sensitivity variables, three types of optimal design prevail for the biogas-integrated system: BPG/PV/Battery, BPG/PV/Wind/Battery, and BPG/Wind/Battery. In contrast, the syngas-integrated option displays only two types of optimal systems: SPG/PV/Battery and SPG/PV/Wind/Battery. All those system types include the biomass generator component (BPG or SPG). However, the biomass generator output of the optimal system depends significantly on the fluctuating renewable energy resources (wind and solar). For the option integrated with biogas, the output of the biogas generator varies from 1,821 kWh/year (1% of total electricity production) in the localities most endowed with both resources to 10,660 kWh/year (7% of total electricity production) in the less affluent areas. Similarly, it varies from 10,625 kWh (6%) to 40,502 kWh/year (28%) for the syngas-integrated-option.

The sensitivity analysis results for the solar PV capital cost on the COE and biomass generator production are shown in Figure 18. Lowering solar PV capital cost would substantially reduce COE and biomass power for the biogas-integrated option. A 40% capital cost reduction of solar PV will decrease the COE by 19% and the biogas generator output by 34%. In contrast, although the COE will be reduced by 18% for the syngas-integrated option, the syngas generator output will remain constant.

The sensitivity analysis results of the biomass resource price on the COE and biomass power output shown in Figure 19 indicate distinct effects for both systems. An increase of $30 per unit cost of the feedstock (livestock manure) will increase COE by 15% and decrease the biogas generator output by 77%. In contrast, the same increase in the feedstock price of the syngas generator will result in a rise in the COE of only1% and no change in the generator output.

3.3 | Discussion

Without incorporating biomass energy into the proposed distributed hybrid renewable system, the COE would be

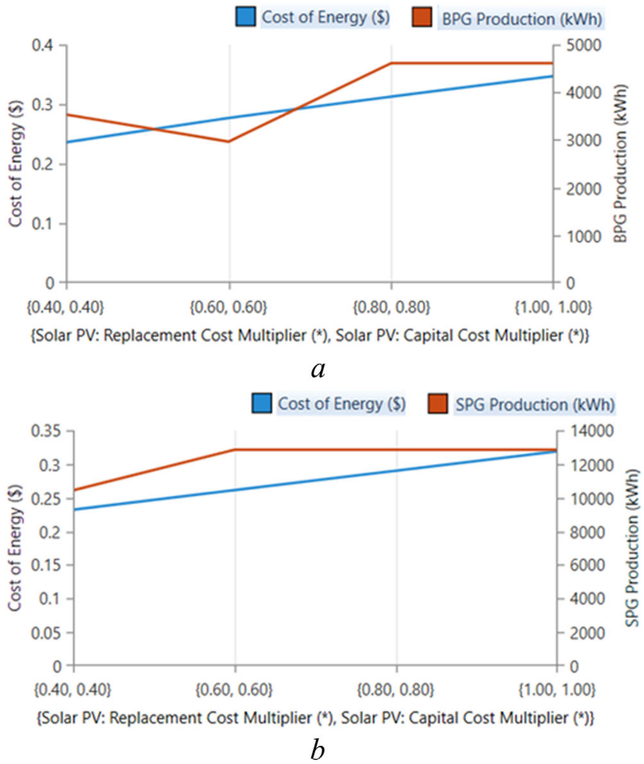

FIGURE 18 Sensitivity analysis results Solar PV capital cost. (a) Anaerobic digester integrated option, (b) Gasifier-integrated option

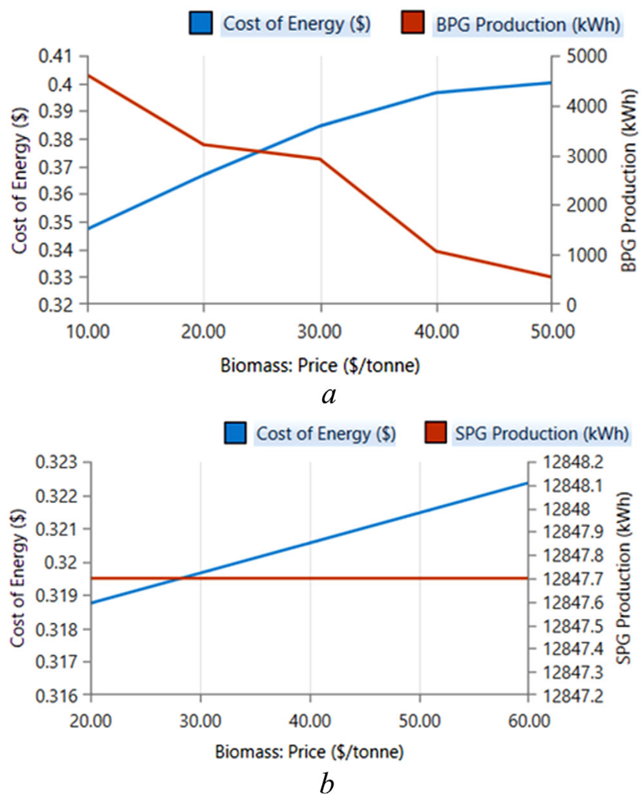

FIGURE 19 Sensitivity analysis results of feedstock price. (a) Anaerobic digester integrated option, (b) Gasifier-integrated option

\$0.447/kWh (the optimal design of category three and four for the biogas and syngas integrated options, respectively). Thus, integrating the biogas (resp. syngas) generator into the proposed distributed hybrid system significantly reduces the COE by up to 29% (resp. 40%), improving the affordability of generated power for developing countries' poor populations. On the other hand, these findings indicate that the SPG can supply the study area with 100% biomass power (category-six optimal system for the gasifier integrated option). However, in that case, the COE would be \$0.546/kWh, or 20% and 70% higher, respectively, than the optimal system without a biomass power generator (category 4) and the overall optimal system (category 1). Such a high COE demonstrates biomass power generators' inability to address SSA's rural electrification challenges as single-power sources.

The findings of this study are consistent with previous research about the effectiveness of incorporating bio-power into distributed HRESs for decreasing the COE. However, it should be noted that the bio-power share of the total electricity generated in the two options analysed is lower than in previous research [20–31]. This may be mainly explained by the significant fall in photovoltaic and wind energy costs over the last few years. For example, solar PV and onshore wind costs decreased by 82% and 29%, respectively, between 2010 and 2020, according to IRENA, the International Renewable Energy Agency [49].

Furthermore, based on the sensitivity analysis of wind speed and solar radiation, one can deduce that incorporating biomass power technology into HRESs will result in COE reductions throughout SSA, regardless of the degree of renewable energy resource abundance. However, the share of biopower in generating electricity will decline as low-endowed areas such as Gabon, Congo, and southern Cameroon give way to highly endowed places such as northern Chad, northern Nigeria, and Eritrea. The sensitivity analysis of solar PV capital costs indicates that the COE reduction function of bio-power in HRESs as described above will continue to operate in the future, independently of the subsequent decline in solar PV energy costs expected by IRENA.

Besides 3,960 tons (resp. 441 tons) of feedstock available per year, only 342 tons (resp. 11.3 tons) or 9% (resp. 2.5%) were used for power generation in the anaerobic digester (resp. gasifier) integrated system. These proportions of feedstock seem to be feasible, given that, in rural areas, biomass resources are also used for other existential needs such as cooking and heating. The average daily amount of biogas required for cooking per person in rural areas, as estimated by Meena and cited by Shahzad et al. [50], is 0.227 m^3/person/day. Considering a gas yield per kg of livestock manure of 0.036 m^3/kg [51], we found that the anaerobic digester's remaining biogas could help meet the cooking needs of 154 people, representing more than 25% of the Babadam population. This will constitute an outstanding contribution towards universal access to clean cooking. In 2018, 905 million people in SSA (83% of the population) lacked access to clean cooking.

Finally, the sensitivity analysis of feedstock prices shows that a rise in livestock manure transportation costs dramatically reduces the utility of incorporating biogas power generation into the anaerobic digester-integrated option. In comparison, this increased transport cost would not affect the share of biopower in the gasifier-integrated alternative, notwithstanding the increased COE.

By integrating biomass power technology into distributed HRESs, rural communities would generate electricity more affordably and without diesel generators. As a result, it will decrease sulphur emissions (which contribute to acid rain), reduce greenhouse gas emissions and dependence on fossil fuels. All of which will benefit the environment preservation and contribute to the achievement of SDG 13, which calls for the incorporation of climate change mitigation policies into national strategies. Additionally, the availability of electrical power would contribute to creating small and medium-sized enterprises and domestic innovations, which are the aims of SDG 9. In addition, it will help create new jobs, promote sustainable growth and improve rural health [52, 53].

The novelty of this study can be summarised in three points: (1) the use of HOMER software to model and simulate different options of biomass PV/Wind/Battery-integrated HRES for rural electrification; (2) the inclusion of all socio-economic sectors in the evaluation of electricity needs in a remote community of SSA; and (3) the broadening of the use of the sensitivity analysis capability of HOMER software to elucidate various aspects of biomass-based HERSs such as the level of the role of integrating biomass power technologies in HRESs and the relevance of transporting the feedstock far from the HRES project area.

Further research is required to address the study's limitations, the most notable of which are: (1) the study's failure to account for the rise in electricity demand with time as a result of population increase and technical progress; and (2) the study's inability to recognise socio-environmental considerations when determining the optimal system configuration.

4 | CONCLUSION

Using HOMER software, we have successfully modelled and simulated two biomass-based HRES options (anaerobic-integrated and gasifier-integrated) in Babadam, a remote community in northern Cameroon. Moreover, the HOMER software sensitivity analysis capabilities helped elucidate several aspects of the proposed systems. Although the gasifier integrated option performed better, both biomass power technologies were shown to contribute to making electricity generated by HRESs more affordable significantly. Sensitivity analysis has established the validity of these findings in terms of time and space within SSA.

Therefore, incorporating biomass power technologies into distributed HRESs can sustainably contribute to addressing rural electrification challenges in SSA.

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APPENDIX A: ABBREVIATIONS

AFD: Average Feedstock per Day
AGMD: Air-Gap Membrane Distillation System
BGED: Break-Even Grid Extension Distance
BPG: Biogas Power Generator
BPT: Biomass Power Technology
BPT: Biomass Power Technology
CC: Cycle Charging
COA: Country of Application
COE: Cost of Energy
CRF: Capital Recovery Factor
DOD: Depth of Discharge
EPP: Excess Power Production
GR: Gasification Ratio;
HOMER: Hybrid Optimisation of Multiple Energy Resources
HRES: Hybrid Renewable Energy Systems
LF: Load Following
LHV: Lower Heating Value
NPC: Total Net Present Cost
NREL: National Renewable Energy Laboratory
PV: Photovoltaic
RF: Recovery Factor
SDG: Sustainable Development Goal
SPG: Syngas Power Generator
SSA: Sub-Saharan Africa
Surface Meteorology and Solar Energy (SSE)
TEP: Total Electrical Production
TFC: Total Feedstock Consumed
VOC: Volatile Organic Compound

APPENDIX B: APPLIANCES' ENERGY RATING, OPERATION HOURS AND LOAD FOR DIFFERENT SECTORS OF ELECTRICITY CONSUMPTION

TABLE B.1

Load sector	Appliances	Rating (W)	Quantity	Operating hours	Load (kWh/day)
A-domestic	CFL	10	200	17:00-23:00	12
	TV	70	100	17:00-23:00	42
	Radio	15	100	17:00-23:00	9
	Mobile charger	15	100	7:00 – 10:00	4.5
	Fan	35	200	20:00 – 4:00	56
B-commercial					
Shops	CFL	10	2	8:00-17:00	0.18
	Fan	35	2	12:00-17:00	0.35
	Refrigerator	600	1	8:00-17:00	5.4
Mini dairy		2500	1	12:00-16:00	10
Flour mill		4500	1	12:00-16:00	18
C-agricultural					
	Cutting machine	1,300	4	12:00-16:00	20.8
	Threshing machine	4200	4	12:00-16:00	67.2
	Pedestal fan	40	4	12:00-16:00	0.64
D-community					
School	CFL	15	10	8:00-15:00	1.05
	Fan	35	2	12:00-15:00	0.21
Health centre	CFL	10	5	17:00-6:00	0.65
	Fan	35	4	12:00-6:00	2.52
	Refrigerator	600	1	00:00-0:00	14.4
Street lights	CFL	100	20	18:00-5:00	22
Total primary daily load (kWh/day)					286.9